The Constrained Dirichlet–Neumann Exchange for Schwarz Coupling in Solid Dynamics

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Abstract

This note specifies a Schwarz coupling for dynamic solid mechanics whose discrete interface work vanishes at the converged state, so that the coupled problem conserves the energy that its time integrators conserve. The method is a Dirichlet–Neumann exchange between two nonoverlapping subdomains, each advanced by its own Newmark scheme or by central difference, on meshes that need not match at the interface and with time steps that may differ by an integer ratio. It differs from the classical Dirichlet–Neumann exchange in five respects: the Neumann side receives the full dynamic (d'Alembert) reaction of its partner; both interface transfers are built from one cross-mass matrix, so that the force transfer is the transpose of the velocity transfer; the Dirichlet side receives the velocity only and derives its displacement and acceleration from its own time integrator; the iteration stops on the interface velocity jump and the interface force residual; and the initial acceleration is computed on the coupled system. At the fixed point the exchange solves the velocity-constrained coupled problem of Gravouil and Combescure, of Prakash and Hjelmstad, and of Connors, Owen, Kuberry, and Bochev, for which the pseudo-energy (the energy functional applied to the velocity and the acceleration) is conserved exactly for linear elasticity, for any pair of Newmark schemes, for any mesh ratio, and, with the coarse-step side as the Dirichlet side, for any integer step ratio. The note states the formulation and its proof, the algorithm in the form needed for an implementation in a production code (interface operators, the exchange, the stopping criterion, the coupled initial acceleration, subcycling, the assignment of the Dirichlet role, relaxation by Aitken or Anderson acceleration, and a direct interface solve for two explicit subdomains), and its verification on a cantilever beam and on nested cylinders: the matched energy is constant to the Schwarz tolerance over 10^4 steps in every configuration measured, and to roundoff with the direct solve, while the classical exchange inverts an element within 1 ms with implicit subdomains.

This version replaces the July–September 2026 note, which recorded the transmission conditions measured before this method (overlap Dirichlet and impedance conditions, the paired impedance condition, and the Robin–Robin condition) and their failures; that version remains in the history of the Norma repository.

Contents

1 Problem and requirements **2**

2 Setting and notation **3**

3 The constrained coupled problem **4**

3.1 Formulation 4

3.2 The energy identity 4

4 Algorithm **6**
4.1 Setup . . . 6
4.2 One window . . . 6
4.3 Stopping criterion . . . 6
4.4 Coupled initial acceleration . . . 7
4.5 Assignment of the Dirichlet role . . . 7
4.6 Relaxation and acceleration . . . 7
4.7 Direct interface solve for two explicit subdomains . . . 8
4.8 Requirements on a subdomain code . . . 8

5 Verification **9**
5.1 Benchmarks . . . 9
5.2 Equal steps . . . 9
5.3 Different time steps . . . 11
5.4 Relaxation and acceleration . . . 13
5.5 Direct interface solve . . . 13
5.6 Nonlinear material . . . 14
5.7 Inexact subdomain solves . . . 14
5.8 The classical exchange . . . 15

6 Limitations and open questions **16**

7 Implementation in Norma **17**

1 Problem and requirements

A solid body $\Omega \subset \mathbb{R}^3$ is decomposed into two subdomains Ω_D and Ω_N that share an interface $\Gamma = \partial\Omega_D \cap \partial\Omega_N$ and are integrated in time independently, each with its own mesh, its own time integrator, and its own time step. The subdomains are coupled by a Schwarz iteration: within each time window the two subdomains are solved in turn, exchanging interface data, until the data converge, following Mota, Tezaur, and Phlipot [1]. The coupling must satisfy the following requirements.

1. *Energy conservation.* At the converged state the coupling does no work on the system: the coupled problem conserves the quantity that the subdomain integrators conserve, to the tolerance of the Schwarz iteration.
2. *Nonconforming meshes.* The two meshes need not share nodes on Γ.
3. *Heterogeneous integrators and time steps.* Each subdomain may use an implicit Newmark scheme or central difference, and the time steps may differ by an integer ratio.
4. *Partitioned solution.* Each subdomain is advanced by its own solver; the coupling requires of a subdomain only operations that a structural dynamics code already provides (Sec. 4.8).

The classical Dirichlet–Neumann exchange satisfies the last three requirements but not the first: with implicit subdomains on the beam of Sec. 5 it inverts an element within 1 ms, and with explicit subdomains it conserves the energy only to 2 to 4% (Sec. 5.8). The constrained exchange of this note satisfies all four.

2 Setting and notation

Semi-discrete equations. After finite element discretization in space, subdomain $k \in \{D, N\}$ satisfies

$$\boldsymbol{M}_k\, \boldsymbol{a}_k + \boldsymbol{f}_k^{\mathrm{int}}(\boldsymbol{u}_k) = \boldsymbol{f}_k^{\mathrm{ext}} + \boldsymbol{C}_k^T \boldsymbol{\lambda}_k, \tag{1}$$

with $\boldsymbol{u}_k$, $\boldsymbol{v}_k$, $\boldsymbol{a}_k$ the nodal displacement, velocity, and acceleration, $\boldsymbol{M}_k$ the mass matrix (lumped for central difference), $\boldsymbol{f}_k^{\mathrm{int}}$ the internal force, $\boldsymbol{f}_k^{\mathrm{ext}}$ the external force, $\boldsymbol{C}_k$ the Boolean restriction of the nodal vector to the interface nodes, and $\boldsymbol{\lambda}_k$ the interface force acting on subdomain k. For linear elasticity $\boldsymbol{f}_k^{\mathrm{int}} = \boldsymbol{K}_k \boldsymbol{u}_k$.

Time integrators. Subdomain k is advanced by the Newmark scheme with parameters (β_k, γ_k) and step Δt_k:

$$\boldsymbol{u}^{n+1} = \boldsymbol{u}^n + \Delta t\, \boldsymbol{v}^n + \Delta t^2 \big[(\tfrac{1}{2} - \beta)\, \boldsymbol{a}^n + \beta\, \boldsymbol{a}^{n+1}\big], \qquad \boldsymbol{v}^{n+1} = \boldsymbol{v}^n + \Delta t \big[(1-\gamma)\, \boldsymbol{a}^n + \gamma\, \boldsymbol{a}^{n+1}\big]. \tag{2}$$

The trapezoidal rule ($\beta = 1/4$, $\gamma = 1/2$, implicit, denoted I) and central difference ($\beta = 0$, $\gamma = 1/2$, explicit with a lumped mass, denoted E) are the two schemes used below. With the predictors $\boldsymbol{u}^{\mathrm{pred}} = \boldsymbol{u}^n + \Delta t\, \boldsymbol{v}^n + (\frac{1}{2} - \beta)\Delta t^2 \boldsymbol{a}^n$ and $\boldsymbol{v}^{\mathrm{pred}} = \boldsymbol{v}^n + (1-\gamma)\Delta t\, \boldsymbol{a}^n$, the scheme gives

$$\boldsymbol{a}^{n+1} = \frac{\boldsymbol{v}^{n+1} - \boldsymbol{v}^{\mathrm{pred}}}{\gamma\, \Delta t}, \qquad \boldsymbol{u}^{n+1} = \boldsymbol{u}^{\mathrm{pred}} + \beta\, \Delta t^2\, \boldsymbol{a}^{n+1}, \tag{3}$$

so a prescribed end-of-step velocity determines the acceleration and the displacement of the step.

Roles. The *Dirichlet side D* receives a kinematic datum on its interface nodes; the *Neumann side N* receives a force. A pair of integrators is written with the integrator of one named subdomain first (II, IE, EI, EE).

Interface operators. Let φ_m^D and φ_n^N be the interface traces of the nodal basis functions of the two sides. The cross-mass matrix and the boundary mass matrices are

$$B_{mn} = \int_\Gamma \varphi_m^D\, \varphi_n^N \,\mathrm{d}S, \qquad (\boldsymbol{W}_D)_{ml} = \int_\Gamma \varphi_m^D\, \varphi_l^D \,\mathrm{d}S, \qquad (\boldsymbol{W}_N)_{nl} = \int_\Gamma \varphi_n^N\, \varphi_l^N \,\mathrm{d}S, \tag{4}$$

assembled once on the reference configuration, per Cartesian component. The kinematic transfer to the Dirichlet side and its transpose are

$$\Pi_D = \boldsymbol{W}_D^{-1} \boldsymbol{B}, \qquad \Pi_D^T = \boldsymbol{B}^T \boldsymbol{W}_D^{-1}. \tag{5}$$

Π_D maps the interface values of side N to the L^2 projection of their trace onto the trace space of side D; it reproduces constants when $\boldsymbol{B}\,\mathbf{1} = \boldsymbol{W}_D \mathbf{1}$, that is, when the quadrature of $\boldsymbol{B}$ is exact on products of the two trace spaces.

Windows. The Schwarz controller advances in *stops*. Within a stop each subdomain advances to the stop end in its own steps; the *Schwarz iteration* repeats a solve of each subdomain with interface data exchanged between the solves until the data converge, and the stop is then committed. A stop that spans m steps of a subdomain is a *window* of m substeps for that subdomain.

Energies. The energy is

$$E = \sum_k \Big(\tfrac{1}{2}\, \boldsymbol{v}_k^T \boldsymbol{M}_k\, \boldsymbol{v}_k + \mathrm{SE}(\boldsymbol{u}_k) \Big), \tag{6}$$

with SE the strain energy. For central difference the kinetic term is evaluated in the staggered form $\frac{1}{2}\,\boldsymbol{v}^T\boldsymbol{M}\boldsymbol{v} - \frac{\Delta t^2}{8}\,\boldsymbol{a}^T\boldsymbol{M}\boldsymbol{a}$, which is the kinetic part of the quadratic that the scheme conserves exactly on a linear problem; the consistent-mass kinetic energy at integer steps misrepresents the short-wavelength content of a sharp front by several percent. For linear elasticity the pseudo-energy is

$$\tilde{E} = \sum_k \Big(\tfrac{1}{2}\, \boldsymbol{a}_k^T \boldsymbol{S}_k\, \boldsymbol{a}_k + \tfrac{1}{2}\, \boldsymbol{v}_k^T \boldsymbol{K}_k\, \boldsymbol{v}_k \Big), \qquad \boldsymbol{S}_k = \boldsymbol{M}_k + \Big(\beta_k - \frac{\gamma_k}{2} \Big) \Delta t_k^2\, \boldsymbol{K}_k, \tag{7}$$

the energy functional applied to the velocity and the acceleration, which is the energy of the time-differentiated system $\boldsymbol{M}\dot{\boldsymbol{a}} + \boldsymbol{K}\boldsymbol{v} = \dot{\boldsymbol{f}}$; its units are those of energy divided by time squared. $\boldsymbol{S}_k$ is $\boldsymbol{M}_k$ for the trapezoidal rule and $\boldsymbol{M}_k - \frac{1}{4}\Delta t_k^2 \boldsymbol{K}_k$ for central difference, positive definite exactly below the stability limit of central difference. Relative to E, $\tilde{E}$ weights each vibration mode by the square of its frequency.

3 The constrained coupled problem

3.1 Formulation

The coupled problem consists of (1) on both sides with one interface force acting in opposite directions,

$$\boldsymbol{\lambda}_D = \boldsymbol{\lambda}, \qquad \boldsymbol{\lambda}_N = -\Pi_D^T \boldsymbol{\lambda}, \tag{8}$$

and the velocity constraint at the end of every window,

$$\boldsymbol{v}_D = \Pi_D\, \boldsymbol{v}_N \quad \text{on } \Gamma. \tag{9}$$

The interface force $\boldsymbol{\lambda}$ is the Lagrange multiplier of (9). On matching meshes Π_D is the identity on the interface nodes and the formulation is that of Gravouil and Combescure [2], a dual Schur formulation with velocity continuity. Prakash and Hjelmstad [6] impose the constraint only at the end of a coarse step, with the force on the fine side interpolated linearly in time; Connors, Owen, Kuberry, and Bochev [3] cast the same structure as a discontinuous-Galerkin-in-time multirate framework whose lowest flux order is this scheme.

The force on the interface rows of side D is, by (1), its d'Alembert reaction

$$\boldsymbol{r}_D = \boldsymbol{C}_D \big(\boldsymbol{M}_D\, \boldsymbol{a}_D + \boldsymbol{f}_D^{\mathrm{int}} - \boldsymbol{f}_D^{\mathrm{body}} - \boldsymbol{f}_D^{\mathrm{bdry}} \big) = \boldsymbol{\lambda}, \tag{10}$$

with $\boldsymbol{f}_D^{\mathrm{body}}$ the body force and $\boldsymbol{f}_D^{\mathrm{bdry}}$ the other boundary loads acting on the interface rows. The reaction includes the inertia of the interface nodes; the static reaction $-\boldsymbol{f}^{\mathrm{int}}$ omits a term of the size of the reaction itself (Sec. 5.8).

3.2 The energy identity

Connors et al. (their Eq. (102)) give the energy balance of one side. For side k advanced by m_k Newmark steps with a linear stiffness, an interface force $\boldsymbol{f}_k$ on its interface rows, and no other loads,

$$\tilde{E}_k^{m_k} - \tilde{E}_k^0 + \Big(\gamma_k - \tfrac{1}{2} \Big) \sum_{n=1}^{m_k} (\Delta \boldsymbol{a}_k^n)^T \boldsymbol{S}_k\, \Delta \boldsymbol{a}_k^n = \frac{1}{\Delta t_k} \sum_{n=1}^{m_k} (\Delta \boldsymbol{v}_k^n)^T \boldsymbol{C}_k^T\, \Delta \boldsymbol{f}_k^n, \tag{11}$$

with $\Delta(\cdot)^n = (\cdot)^n - (\cdot)^{n-1}$ and $\tilde{E}_k$ the term of side k in (7). With $\gamma_k = 1/2$ the second term vanishes.

Proposition 1 (Equal steps)**.** *Let both sides take one step Δt per window, with $\gamma = 1/2$ on both, a linear stiffness, no loads on the interface rows other than the interface force, $\boldsymbol{f}_D = \boldsymbol{\lambda}$ and $\boldsymbol{f}_N = -\Pi_D^T \boldsymbol{\lambda}$, and $\boldsymbol{v}_D = \Pi_D \boldsymbol{v}_N$ at every step. Then $\tilde{E}_D + \tilde{E}_N$ is constant.*

Proof. By (11) the change over one step is $\Delta t^{-1}\big[(\Delta \boldsymbol{v}_D)^T \Delta\boldsymbol{\lambda} - (\Delta \boldsymbol{v}_N)^T \Pi_D^T \Delta\boldsymbol{\lambda}\big] = \Delta t^{-1}(\Delta \boldsymbol{v}_D - \Pi_D \Delta \boldsymbol{v}_N)^T \Delta\boldsymbol{\lambda} = 0$. □

The cancellation needs only the transposed force transfer and the one-sided constraint; it holds for any cross-mass matrix $\boldsymbol{B}$, so for any mesh ratio, and for any pair of Newmark schemes with $\gamma = 1/2$. It is the interface work identity of Gravouil and Combescure (their Eq. (49)) written with Π_D in place of the Boolean interface map. Identical integrators at equal steps conserve E as well: the constraint and the common Newmark relations then give $\boldsymbol{u}_D = \Pi_D \boldsymbol{u}_N$ up to the initial data, and the interface power $\boldsymbol{\lambda}^T(\boldsymbol{v}_D - \Pi_D \boldsymbol{v}_N)$ vanishes at the end of every step.

Proposition 2 (Different steps)**.** *Let side D take one step ΔT per window and side N take m steps $\Delta T/m$, under the conditions of Proposition 1 except that $\boldsymbol{v}_D = \Pi_D \boldsymbol{v}_N$ holds only at the two ends of the window, and let the force on N at its substep j be the linear interpolation*

$$\boldsymbol{f}_N^{\,j} = -\Pi_D^T \Big[\big(1 - \tfrac{j}{m}\big)\boldsymbol{\lambda}^0 + \tfrac{j}{m}\,\boldsymbol{\lambda}^1\Big], \qquad j = 0, \ldots, m, \tag{12}$$

of the forces $\boldsymbol{\lambda}^0$ and $\boldsymbol{\lambda}^1$ on D at the window ends. Then $\tilde{E}_D + \tilde{E}_N$ is constant over the window.

Proof. $\Delta \boldsymbol{f}_N^{\,j} = -\Pi_D^T \Delta\boldsymbol{\lambda}/m$ for every substep, with $\Delta\boldsymbol{\lambda} = \boldsymbol{\lambda}^1 - \boldsymbol{\lambda}^0$, so the right side of (11) for N is $\frac{m}{\Delta T}\sum_{j=1}^{m} (\Delta \boldsymbol{v}_N^j)^T(-\Pi_D^T \Delta\boldsymbol{\lambda}/m) = -\Delta T^{-1}\big(\Pi_D(\boldsymbol{v}_N^m - \boldsymbol{v}_N^0)\big)^T \Delta\boldsymbol{\lambda}$, and that of D is $\Delta T^{-1}(\boldsymbol{v}_D^1 - \boldsymbol{v}_D^0)^T \Delta\boldsymbol{\lambda}$. The two cancel when $\boldsymbol{v}_D = \Pi_D \boldsymbol{v}_N$ at both window ends. □

This is Eq. (103) of Connors et al. with Π_D in place of their Boolean map, their condition (108) at flux order $r = 1$, and Eq. (70) of Prakash and Hjelmstad. The force on N at the window start must be the converged force of the previous window, and the loads on the interface rows of D must be constant within the window, because the reaction (10) is formed from their current value.

The coarse step must be the Dirichlet side. The same sums cancel when the fine-step side is the Dirichlet side and meets the linearly interpolated coarse velocity at every substep, the arrangement of Gravouil and Combescure. That arrangement conserves $\tilde{E}$ but lacks the flux projection property of Connors et al. (their Eq. (80)): the impulse that the fine side receives over the window, the trapezoidal sum of its reactions at every substep, differs from the impulse that the coarse side receives, the trapezoid of the two end values. The interface impulse residual of a window,

$$\boldsymbol{\rho} = \sum_{j=1}^{m} \frac{\Delta T}{2m}\big(\boldsymbol{f}_N^{\,j-1} + \boldsymbol{f}_N^{\,j}\big) + \Pi_D^T \frac{\Delta T}{2}\big(\boldsymbol{\lambda}^0 + \boldsymbol{\lambda}^1\big), \tag{13}$$

vanishes identically in the coarse-step arrangement. In the fine-step arrangement it is of the order of the impulse itself, and the energy E varies by 12 to 51% over the period of the first bending mode of the beam while $\tilde{E}$ is conserved (Sec. 5.3). $\tilde{E}$ weights the lowest mode least, so it does not detect the variation.

Scope. Propositions 1 and 2 hold for a linear stiffness. For a nonlinear material the exchange still closes the interface exactly and does no interface work at the fixed point; the energy then varies by the time-step error of the subdomain integrators themselves. For central difference that error is of order Δt^2 and does not grow: the scheme conserves a modified energy that differs from E by a state-dependent term of order Δt^2 [8]. Conservation of E to roundoff for a nonlinear material requires an energy-momentum integrator [9, 10] in each subdomain and a midpoint velocity constraint in the exchange, which this note does not treat.

4 Algorithm

4.1 Setup

1. Assign the Dirichlet role (Sec. 4.5).
2. Assemble $\boldsymbol{B}$, $\boldsymbol{W}_D$, and $\boldsymbol{W}_N$ of (4) on the reference configuration with the facet quadrature of the finer side, and form Π_D of (5). Check that Π_D reproduces constants (the partition-of-unity error $\max|\Pi_D\mathbf{1}-\mathbf{1}|$ is at roundoff when the quadrature is exact). Only Π_D and its transpose enter the exchange; the projection in the other direction is never formed, which matters on curved or non-nested interfaces where it is inaccurate.
3. Compute the initial acceleration on the coupled system (Sec. 4.4).

4.2 One window

Given the converged state at the window start and the force $\boldsymbol{\lambda}^0$ of the previous window:

1. Initialize the datum $\boldsymbol{v}_D^*$ of the Dirichlet side, for example with the projected partner velocity of the previous window.
2. Repeat until the stopping criterion of Sec. 4.3 holds:
   (a) Advance side D over the window with $\boldsymbol{v}_D = \boldsymbol{v}_D^*$ imposed on its interface nodes at the window end, the interface acceleration and displacement from (3), and form its reaction $\boldsymbol{\lambda}^1 = \boldsymbol{r}_D$ by (10).
   (b) Advance side N over its substeps with the force (12) on its interface nodes (at equal steps, $\boldsymbol{f}_N = -\Pi_D^T\boldsymbol{\lambda}^1$).
   (c) Form the new datum $\Pi_D\boldsymbol{v}_N$ at the window end and relax it (Sec. 4.6) into $\boldsymbol{v}_D^*$.
3. Commit the window and keep $\boldsymbol{\lambda}^1$ as the next $\boldsymbol{\lambda}^0$.

Each Schwarz iteration restores both sides to the state at the window start before it advances them. With the displacement constraint instead of the velocity constraint (trapezoidal pairs only) the datum is $\Pi_D\boldsymbol{u}_N$ and the velocity and acceleration follow from the Newmark relations; the velocity constraint is the general choice.

4.3 Stopping criterion

The iteration stops when both the one-sided relative velocity jump and the relative force residual fall below a tolerance ϵ:

$$\frac{\|\boldsymbol{v}_D - \Pi_D\boldsymbol{v}_N\|_{\boldsymbol{W}_D}}{\|\boldsymbol{v}_D\|_{\boldsymbol{W}_D}} < \epsilon, \qquad \frac{\|\boldsymbol{f}_N + \Pi_D^T\boldsymbol{r}_D\|_{\boldsymbol{W}_N^{-1}}}{\|\Pi_D^T\boldsymbol{r}_D\|_{\boldsymbol{W}_N^{-1}}} < \epsilon, \tag{14}$$

the force residual taken at every substep of the window. The norms are the discrete L^2 norms of the trace and of its dual. An absolute floor, if any, must be given in the units of the constrained quantity; a displacement tolerance divided by the step is not a valid velocity floor (on the cylinders of Sec. 5 it amounts to 200 m/s and stops the iteration after one pass). A stop at which the residual stalls above ϵ for three consecutive iterations without falling 5% below its smallest value may be accepted with a warning; the residual is not monotone under Aitken or Anderson acceleration, so a single non-decreasing iteration is not a stall. Every stopping tolerance leaves a drift of the conserved quantity of the order of ϵ per window, accumulated over the windows of the run (Sec. 5.4).

4.4 Coupled initial acceleration

The initial accelerations must satisfy the coupled problem at $t = 0$, including the acceleration constraint $\boldsymbol{a}_D = \Pi_D \boldsymbol{a}_N$, or the first step changes $\tilde{E}$ by 0.2 to 0.9% and, for an explicit pair, the difference of the two interface accelerations alternates in sign at every step without decaying. The coupled initial acceleration is a Schwarz iteration at $t = 0$ in which D imposes the projected partner acceleration, N receives the d'Alembert reaction of D, and each side solves for its acceleration with the interface rows included; it converges in 2 iterations on conforming meshes with identical integrators, 16 to 24 on nonconforming meshes, and about 30 for mixed pairs at a fixed relaxation factor of 0.5, and in 2 to 9 with Anderson acceleration. For two explicit sides the direct solve of Sec. 4.7 gives it in one step.

4.5 Assignment of the Dirichlet role

The linearized gain of one Schwarz iteration at fixed relaxation is

$$\boldsymbol{G} = -\,\Pi_D\, \boldsymbol{H}_N\, \Pi_D^T\, \boldsymbol{S}_D, \tag{15}$$

with $\boldsymbol{H}_N = \boldsymbol{C}_N \tilde{\boldsymbol{M}}_N^{-1} \boldsymbol{C}_N^T$ the interface flexibility of N ($\tilde{\boldsymbol{M}} = \boldsymbol{M} + \beta \Delta t^2 \boldsymbol{K}$, the lumped mass for central difference) and $\boldsymbol{S}_D$ the interface dynamic stiffness of D. A fixed relaxation factor θ converges if and only if the spectral radius of $(1-\theta)\boldsymbol{I} + \theta \boldsymbol{G}$ is below one; for a real spectrum in $[a, b]$ the best factor is $\theta^* = 2/((1-a) + (1-b))$ with contraction $(b-a)/((1-a) + (1-b))$. Table 2 lists the computed eigenvalues on the beam. The rules that follow, in order of precedence:

1. With different time steps, the coarse-step side is the Dirichlet side (Proposition 2 and the impulse residual).
2. With different integrators, the implicit side is the Dirichlet side: with the explicit side as the Dirichlet side the eigenvalues reach -13.3 and every factor $\theta \geq 0.5$ diverges.
3. With the same integrator on a nonconforming interface, the finer mesh is the Dirichlet side: the eigenvalues then lie in $[g, 0]$ with $|g| \leq 1$; with the coarser mesh as the Dirichlet side they reach -8 for an explicit pair.

When rules 1 and 3 conflict (a finer mesh with a finer step), giving the finer mesh the coarser step satisfies both where the stability limit of an explicit side allows it. With Anderson acceleration every assignment measured converges, so the rules then set the cost, not the feasibility.

4.6 Relaxation and acceleration

Aitken. The Aitken factor must be formed from the projected interface trace $\Pi_D \boldsymbol{v}_N$ that D receives. Formed from the whole partner field it tends to 1: the interior values do not enter the exchange, follow a map of zero gain, and dominate the residual, and at $\theta = 1$ the interface map, of gain near -1, does not contract.

Anderson. Anderson acceleration [4], in the interface quasi-Newton form of Degroote, Bathe, and Vierendeels [5], combines the last m iterates. With $\boldsymbol{x}_k$ the datum D used, $\boldsymbol{f}_k = G(\boldsymbol{x}_k) - \boldsymbol{x}_k$ the residual, $\boldsymbol{t}_k$ its projected trace, and $\Delta\boldsymbol{T}$, $\Delta\boldsymbol{X}$, $\Delta\boldsymbol{F}$ the differences of the last $m+1$ iterates,

$$\boldsymbol{\gamma} = \arg\min \|\boldsymbol{t}_k - \Delta\boldsymbol{T}\boldsymbol{\gamma}\|, \qquad \boldsymbol{x}_{k+1} = \boldsymbol{x}_k + \beta\,\boldsymbol{f}_k - (\Delta\boldsymbol{X} + \beta\,\Delta\boldsymbol{F})\,\boldsymbol{\gamma}, \tag{16}$$

with mixing parameter β, the least-squares problem solved by QR with the oldest column dropped while the condition number exceeds 10^{10}. The first iteration of a window has no history and is the fixed relaxation $\boldsymbol{x}_1 = \boldsymbol{x}_0 + \beta\boldsymbol{f}_0$; for $k \geq 1$ the term $\beta\boldsymbol{f}_k$ is damped by β and the history term corrects it. With $\beta = 1$ the update is the undamped quasi-Newton step. On the beam (1 ms, $\epsilon = 10^{-12}$), $\beta = 0.5$ needs 7.0, 23.1, and 7.9 iterations per window on the implicit pair, on the mixed pair with the explicit side as the Dirichlet side, and on the explicit pair at ratio 1:0.5 with 4:1 steps and the coarse mesh as the Dirichlet side, against 8.9, 28.7, and 8.8 at $\beta = 1$, with the same energies; $\beta = 0.5$ is the recommended value and the default in Norma. For a linear interface map with $\beta = 1$ and a depth at least the iteration count, the iterate equals the map applied to the GMRES iterate of the same index (Walker and Ni, Theorem 2.2). The first iteration of a window has no history and is not entered in it. A depth of 10 covers every case measured; a depth of 5 is too short for an explicit Dirichlet side, whose window needs 12 to 16 iterations.

State. The relaxation state is kept per interface pair and per substep time, and cleared at every stop. A state shared across substeps blends iterates of different times, which acts as a causal filter on the exchanged datum and moves the converged solution. Aitken and Anderson acceleration apply to stops of one substep; a stop that spans several substeps of the relaxed side uses a fixed factor.

4.7 Direct interface solve for two explicit subdomains

For central difference on both sides at equal steps the iteration can be replaced by one solve per step. Central difference computes $\boldsymbol{u}^{n+1}$ before the acceleration, and the internal force at $\boldsymbol{u}^{n+1}$ does not depend on the interface force. Advancing both sides with free interface rows and no interface force gives the velocities $\boldsymbol{v}^{\text{free}}$; the interface force adds $\gamma\Delta t\,\boldsymbol{m}_D^{-1}\boldsymbol{\lambda}$ to the velocity of D and $-\gamma\Delta t\,\boldsymbol{m}_N^{-1}\Pi_D^T\boldsymbol{\lambda}$ to that of N, with $\boldsymbol{m}_D$, $\boldsymbol{m}_N$ the lumped masses of the interface rows. The velocity constraint then gives, per Cartesian component,

$$\gamma\Delta t\big(\boldsymbol{m}_D^{-1} + \Pi_D\,\boldsymbol{m}_N^{-1}\,\Pi_D^T\big)\boldsymbol{\lambda} = \Pi_D\boldsymbol{v}_N^{\text{free}} - \boldsymbol{v}_D^{\text{free}}. \tag{17}$$

The matrix is symmetric positive definite, of the size of the Dirichlet trace, and constant while the lumped masses and Π_D are; rows held by other Dirichlet conditions take a zero inverse mass. It is the interface operator $\boldsymbol{H} = \sum_k \gamma_k\Delta t_k\boldsymbol{C}_k\tilde{\boldsymbol{M}}_k^{-1}\boldsymbol{C}_k^T$ of Gravouil and Combescure for two lumped-mass sides, written on the Dirichlet trace. Factor it once (a dense Cholesky factorization of 3920 unknowns per component on the cylinders takes 0.7 s), and at every step advance both sides, solve (17), and correct the interface accelerations and velocities. The solve is exact for nonlinear materials, because the internal force is evaluated at the predicted displacement. The same matrix gives the coupled initial acceleration from $\boldsymbol{a}_D = \Pi_D\boldsymbol{a}_N$.

4.8 Requirements on a subdomain code

The exchange requires of each subdomain code, beyond a standard step:

1. a step with a prescribed velocity on a set of boundary nodes, with the displacement and acceleration of those nodes from (3) (the Dirichlet side);
2. a step with a prescribed nodal force on a set of boundary nodes, constant or linear in time over the window (the Neumann side);
3. the nodal d'Alembert reaction (10) on a set of boundary nodes after a step;
4. saving and restoring the state at the window start;
5. the boundary mass matrix and the facet quadrature of its interface, to assemble $\boldsymbol{B}$ (performed once, by either code or by the coupler).

An implicit code provides items 1 to 3 as a Newton solve at fixed boundary data; an explicit code provides them as a division by the lumped mass. The coupler itself holds Π_D, the relaxation state, and, for two explicit sides, the factorization of (17).

5 Verification

5.1 Benchmarks

Cantilever beam. A linear elastic bar ($E = 6.895\,\mathrm{GPa}$, $\nu = 0.25$, $\rho = 2768\,\mathrm{kg/m^3}$; $254 \times 25.4 \times 12.7\,\mathrm{mm}$), clamped at $x = 0$ and released at rest from the bent shape $u_y = 0.3937\,x^2$ (tip deflection 25.4 mm), with initial energy 1506 J and a first bending period of about 10 ms. It is split at $x = 127\,\mathrm{mm}$ into a clamped part and a free part. The free part is meshed with HEX8 elements of 6.35 mm; the clamped part at 6.35 mm (conforming), 8.47 mm (ratio 1:0.75, facets not nested), or 12.7 mm (ratio 1:0.5, nested). Stops of $1\,\mu$s (or $0.5\,\mu$s where stated), 10 ms. The first letter of an integrator pair is the clamped part.

Nested cylinders. A steel shell (outer radius 88.9 mm, height 330 mm), a soft filler of one fiftieth of the shell's wave impedance, and an offset core, all neohookean, launched at $-35\,\mathrm{m/s}$ against a support that fixes the bottom face in z. The interface is a closed surface inside the filler around the core (a lateral surface and two caps); the inner mesh is 5.1 mm and the outer 6.35 mm (ratio 1:0.8, not nested), 17 to 26 thousand elements per subdomain. Central difference on both sides at a Courant number of 0.5, stops of $0.5\,\mu$s, 2 ms. A linear elastic version with the same meshes (filler $E = 0.214\,\mathrm{GPa}$, $\nu = 0.494$) moves at $-5\,\mathrm{m/s}$ for 1 ms. The inner side is the Dirichlet side.

Ground truth. The undecomposed beam keeps E and $\tilde{E}$ within 1.7×10^{-13} over 10 ms with the trapezoidal rule and within 2.7×10^{-14} with central difference; the trapezoidal value requires a direct linear solver and a tight tolerance on the initial-acceleration solve.

5.2 Equal steps

Figure 1 shows the energy of the conforming beam for the classical and the constrained exchange. Table 1 gives the largest changes of E and $\tilde{E}$ over 10 ms at a Schwarz tolerance of 10^{-12} with the coupled initial acceleration.

The matched quantity is constant to between 10^{-14} and 4×10^{-10} in every case: $\tilde{E}$ for every pair, and E as well for identical integrators. The residual is the tolerance accumulated over 10^4 stops; the explicit pair at $\theta = 0.5$ converges in one iteration exactly. For the mixed pairs E varies by 0.1 to 0.4%, which is the expected behavior of the velocity constraint between different

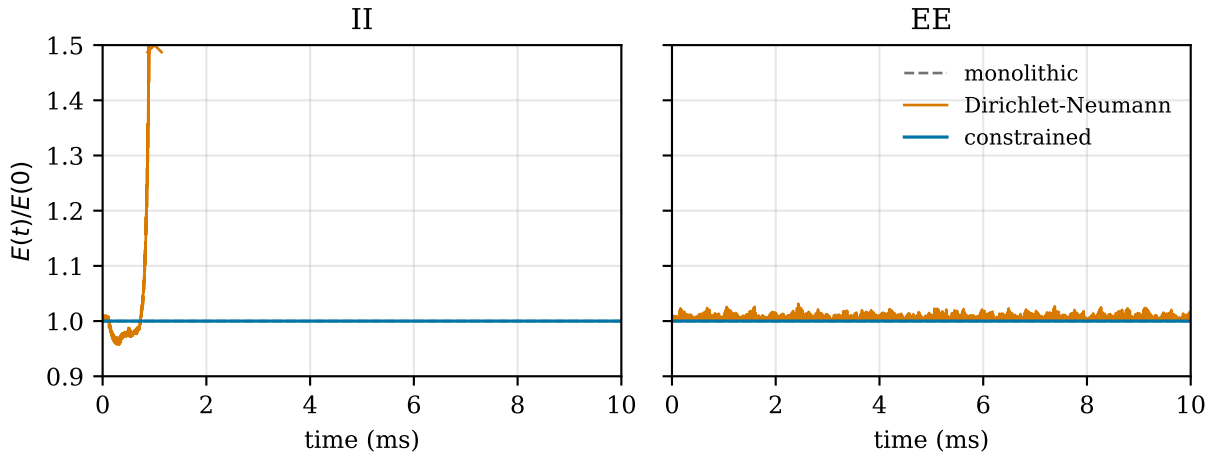


Figure 1: Energy of the conforming beam, stops of 1 μs, 10 ms: the monolithic reference, the classical Dirichlet–Neumann exchange (Schwarz tolerance 10^{-8} on the displacement update, recursive Aitken), and the constrained exchange (tolerance 10^{-12}, Anderson acceleration). The classical exchange inverts an element at 1.0 ms with implicit subdomains (cross); the constrained exchange coincides with the reference.

Table 1: Constrained exchange on the beam at equal steps (1 μs, 10 ms, tolerance 10^{-12}, fixed relaxation 0.5, coupled initial acceleration, velocity constraint unless marked u). δE and $\delta\tilde{E}$ are the largest $|E(t)/E(0) - 1|$ and $|\tilde{E}(t)/\tilde{E}(0) - 1|$; jump is the largest relative one-sided interface displacement jump over the last 100 stops; it. is the mean and maximum number of Schwarz iterations per stop. D is the Dirichlet side: the free part, or the implicit side of a mixed pair.

meshes	pair	δE	$\delta\tilde{E}$	jump	it.	wall (s)
conforming	II, u	2.2e−11	5.3e−11	5.4e−13	7.1/8	215
conforming	II	2.8e−11	7.0e−11	1.7e−12	7.0/8	206
conforming	EE	1.2e−14	3.6e−14	7.5e−15	1.0/1	46
conforming	IE	1.2e−3	6.3e−12	9.6e−4	29.0/32	417
conforming	EI	3.7e−3	4.2e−10	9.3e−4	27.1/28	419
1:0.5	II, u	1.2e−10	4.3e−11	9.8e−13	17.3/19	437
1:0.5	II	1.2e−10	1.8e−11	5.6e−12	17.2/18	436
1:0.5	EE	3.5e−11	6.9e−11	1.7e−11	23.7/26	169
1:0.75	II	1.7e−11	5.2e−11	1.5e−12	13.5/14	350
1:0.75	EE	2.2e−11	6.1e−11	6.0e−12	16.2/17	157

integrators, and the root mean square displacement jump stays near 10^{-5} m (0.1 to 0.2% of the initial interface displacement), growing by at most 10% over the 10 ms, the displacement drift that velocity continuity permits [7]. At 1:0.75 the projector onto the coarse side reproduces constants only to 1.2×10^{-2}; the exchange is unaffected because only Π_D, whose error is 7×10^{-16}, enters it.

The iteration counts of Table 1 were measured with the stopping rule of September 30, 2026, which also accepted a stop when the root mean square jump of the constrained velocity fell below the controller's absolute tolerance divided by the step. Without this floor, removed on October 1, 2026 (commit `d96ab13f`), the conforming II pair with the inputs of `studies/dynamic-stability` (case `beam_no-cd_II_L1_r100_fixed`, absolute tolerance 10^{-15}) takes 8.85 Schwarz iterations per stop (maximum 10) instead of 7.97 (maximum 8) with the floor, with $\delta E = 7.3 \times 10^{-13}$ and $\delta \tilde{E} = 1.3 \times 10^{-12}$.

Table 2 gives the gain (15) on which the role rules of Sec. 4.5 rest.

Table 2: Eigenvalues of the gain (15) at $\Delta t = 10^{-6}$ s and the spectral radius ρ of the relaxed iteration. Conforming: the meshes of the cantilever example; 1:0.5 and 1:0.75: free part 6.35 mm, clamped part 12.7 and 8.47 mm. D is the Dirichlet side.

meshes	pair, D	eigenvalues	$\rho(0.5)$	$\rho(0.75)$	$\rho(1)$
conforming	II, either	$[-1.09, -0.92]$	0.05	0.57	1.09
conforming	EE, either	-1	0	0.50	1.00
conforming	IE or EI, D explicit	$[-13.3, -1.63]$	6.1	9.7	13.3
conforming	IE or EI, D implicit	$[-0.61, -0.075]$	0.46	0.21	0.61
1:0.5	II, D fine	$[-0.55, 0]$	0.50	0.25	0.55
1:0.5	II, D coarse	$[-1.99, -1.83]$	0.50	1.24	1.99
1:0.5	EE, D fine	$[-0.50, 0]$	0.50	0.25	0.50
1:0.5	EE, D coarse	$[-8.0, -2.0]$	3.5	5.8	8.0
1:0.75	II, D fine	$[-0.82, 0]$	0.50	0.37	0.82
1:0.75	EE, D fine	$[-0.75, 0]$	0.50	0.31	0.75

5.3 Different time steps

The beam with the free part subcycled 4:1 (coarse step 1 μs on the clamped part, fine step 0.25 μs on the free part) was run with each side as the Dirichlet side, and on the 1:0.5 meshes in two assignments: (A) the clamped part, coarse in mesh and step, as the Dirichlet side; (B) the free part, fine in mesh, given the coarse step and the Dirichlet role, with the clamped part subcycled. Table 3 gives the results at a fixed relaxation of 0.5; Fig. 2 shows the energy of the conforming runs under Anderson acceleration.

With the coarse-step side as the Dirichlet side, $\tilde{E}$ is constant to between 4.5×10^{-13} and 3.7×10^{-12} and the impulse residual (13) is at roundoff. With the fine-step side as the Dirichlet side, $\tilde{E}$ changes by 2 to 4×10^{-9} (the accumulated stopping error, which falls with the tolerance), the relative impulse residual has a median of 0.67 (II) and 0.95 (EE), its net value in the bending direction sums to -17 and -29 N s over the run while the other components stay below 5×10^{-12} N s, and E swings by 20 and 51%. Assignment (A) with explicit sides has gain eigenvalues in $[-7.9, -2.0]$; it diverges at every fixed factor from 0.25 up, already in the coupled initial acceleration, and converges with Anderson acceleration (Table 4). Assignment (B) satisfies both role rules and has gain eigenvalues in $[-0.56, 0]$.

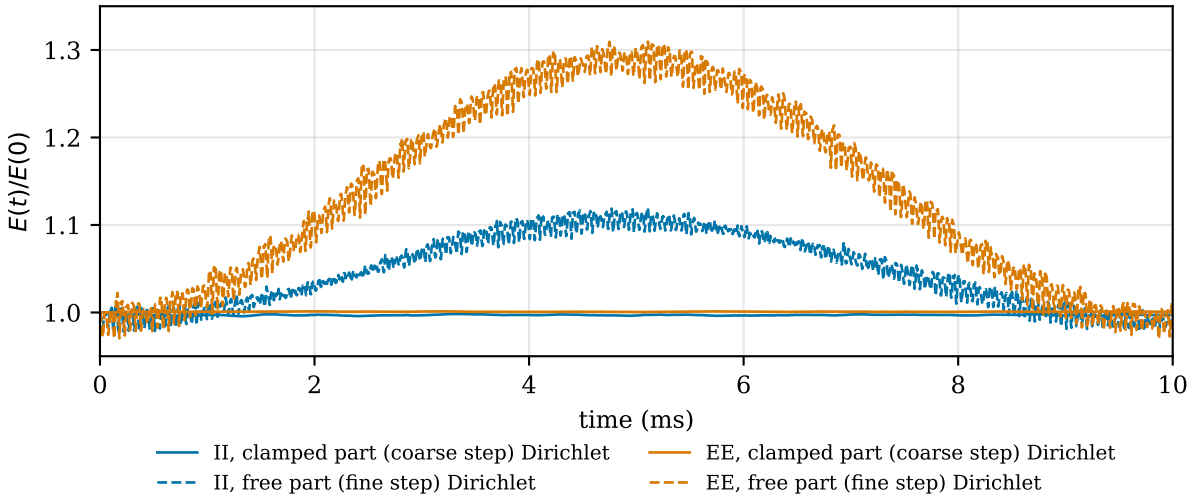


Figure 2: Energy of the conforming beam with the free part subcycled 4:1, tolerance 10^{-12}, Anderson acceleration. With the clamped part (coarse step) as the Dirichlet side the energy stays within 0.4%; with the free part (fine step) as the Dirichlet side it rises by 12% (II) and 31% (EE) and returns over the first bending period, while $\tilde{E}$ stays constant to 10^{-9} in both arrangements.

Table 3: Beam with one side subcycled 4:1, 10 ms, $\theta = 0.5$, relative tolerance 10^{-12}, no absolute tolerance. D: the Dirichlet side; "coarse" and "fine" refer to the step. $E(t)/E(0)$ at 1, 2, 5, and 10 ms; $\delta E = \max_t |E/E(0) - 1|$ with the sign of the final value; drift: root mean square interface displacement jump at 10 ms (m); impulse: largest net residual per window in y (N s) and the median relative residual; it.: Schwarz iterations per stop, mean/maximum (each iteration advances the fine side four substeps); wall time in seconds on four threads. The 1:0.5 (A) run with explicit sides ended at the first stop. Lower part: the spectral radius ρ of the relaxed iteration from the linearized gain at a stop.

meshes	pair, D	1	2	5	10	δE	$\delta\tilde{E}$	drift	impulse y, rel.	it.	wall
conf.	EE, D coarse	1.0005	1.0009	1.0003	1.0006	+1.1e−3	−4.5e−13	3.1e−6	1.9e−15, 1.1e−13	6.0/7	139
conf.	EE, D fine	1.0432	1.1376	1.4698	0.9938	−5.1e−1	+4.4e−9	4.2e−6	4.8e−2, 0.95	36.6/42	564
conf.	II, D coarse	0.9968	0.9982	0.9972	0.9971	−3.8e−3	+1.6e−12	3.1e−6	2.6e−14, 2.2e−13	10.1/15	560
conf.	II, D fine	1.0168	1.0682	1.1977	0.9872	−2.0e−1	+2.0e−9	5.5e−14	4.9e−2, 0.67	35.7/40	1763
1:0.5	EE, (A)	diverged in the coupled initial acceleration; ended at the first stop									
1:0.5	EE, (B)	0.9994	0.9995	0.9993	0.9993	−8.9e−4	+3.7e−12	1.5e−6	2.8e−17, 1.9e−16	26.9/31	315
1:0.5	II, (A)	0.9989	0.9992	0.9984	0.9993	−1.9e−3	−4.7e−13	2.2e−6	5.9e−14, 4.2e−13	44.5/54	2555
1:0.5	II, (B)	1.0008	1.0009	1.0008	1.0008	+1.7e−3	+1.6e−12	1.6e−6	2.4e−17, 1.9e−16	18.8/21	829

meshes	pair	Dirichlet side	$\rho(0.25)$	$\rho(0.5)$	$\rho(0.75)$	$\rho(1)$
conforming	EE	clamped, coarse step	0.50	0.008	0.50	1.00
conforming	II	clamped, coarse step	0.50	0.083	0.63	1.17
1:0.5	EE	(A) clamped, coarse mesh and step	1.23	3.47	5.70	7.93
1:0.5	II	(A) clamped, coarse mesh and step	0.26	0.55	1.32	2.09
1:0.5	EE	(B) free, fine mesh, coarse step	0.75	0.50	0.25	0.50
1:0.5	II	(B) free, fine mesh, coarse step	0.75	0.50	0.25	0.56

5.4 Relaxation and acceleration

Table 4 gives the Schwarz iterations per stop under the three relaxation schemes at a tolerance of 10^{-12}. Aitken acceleration settles at factors near θ^* of Sec. 4.5: 0.33 on (A) implicit, 0.16 to 0.33 on (A) explicit, 0.64 to 0.84 on (B), and 0.07 to 0.21 with an explicit Dirichlet side. Anderson acceleration at depth 10 needs fewer iterations than Aitken in every case, converges where every fixed factor diverged, and with an explicit Dirichlet side removes the drift of $\tilde{E}$ that Aitken left through stops accepted at the floor of the force residual (6.8×10^{-11} against 6.5×10^{-14}). The energies agree across the schemes to the digits measured.

Table 4: Schwarz iterations per stop (mean/maximum) and iterations of the coupled initial acceleration, beam, 10 ms, tolerance 10^{-12}. Equal steps at 0.5 μs on conforming meshes; (A) and (B) as in Sec. 5.3 at 1 μs with 4:1 subcycling; EI and IE with the explicit side as the Dirichlet side at equal steps.

case	fixed 0.5	Aitken	Anderson, $m = 5$	Anderson, $m = 10$	initial, Aitken	initial, Anderson
equal steps, II	5.9/6	5.8/6	5.0/6	5.0/6	3	2
equal steps, EE	1.0/1	1.0/1	–	–	–	–
(A) II	44.5/54	8.8/15	6.0/8	6.0/8	7	4
(A) EE	diverged	20.9/87	8.9/9	8.9/9	21	7
(B) II	18.8/21	7.0/9	5.0/7	5.0/7	5	3
(B) EE	26.9/31	10.9/13	5.0/5	5.0/5	15	4
EI, D explicit	diverged	34.6/45	69.2/118	16.0/16	47	9
IE, D explicit	diverged	33.9/46	53.6/108	13.9/14	47	8

5.5 Direct interface solve

Table 5 compares the direct solve (17) with the iteration and with the monolithic runs. It conserves E and $\tilde{E}$ to roundoff on the beam at every mesh ratio and on the linear cylinders, at a lower cost than the iteration on nonconforming meshes; on the cylinders the relative interface velocity jump at the end of every stop is at most 7×10^{-16}. On the linear cylinders $\tilde{E}$ varies more than E because the deformation gradient is formed from the current coordinates $\boldsymbol{X} + \boldsymbol{u}$, which loses digits in proportion to $|\boldsymbol{X}|/|\boldsymbol{u}|$; its per-step increments have a standard deviation of 1.1×10^{-13} and no significant mean.

Table 5: Direct interface solve, two explicit sides at equal steps. Beam: 10 ms, 1 μs; cylinders: 1 ms, 0.5 μs, eight threads. δE and $\delta\tilde{E}$ as in Table 1; drift is the root mean square interface displacement jump at the end.

case	exchange	δE	$\delta\tilde{E}$	drift (m)	it.	wall (s)	tolerance
beam, conforming, EE	direct	1.5e−14	4.4e−14	1.6e−17	–	39	–
	iterated	1.2e−14	3.6e−14	4.8e−17	1.0/1	45	rel. 10^{-12}
beam, 1:0.5, EE	direct	7.8e−15	4.5e−14	1.8e−17	–	39	–
	iterated	2.2e−12	4.4e−12	4.5e−15	26.8/30	176	rel. 10^{-12}
beam, 1:0.75, EE	direct	1.2e−14	6.0e−14	2.7e−17	–	39	–
	iterated	1.6e−12	4.6e−12	2.3e−15	18.1/20	164	rel. 10^{-12}
linear cylinders, −5 m/s	monolithic	1.3e−13	6.2e−12	–	–	371	–
	direct	9.9e−14	3.2e−12	6.0e−19	–	735	–
linear cylinders, −35 m/s	monolithic	1.2e−14	4.8e−13	–	–	359	–
	direct	1.1e−14	4.3e−13	4.3e−18	–	731	–

5.6 Nonlinear material

On the neohookean cylinders the direct solve closes the interface to 7×10^{-16} at every step, and E stays within 6.4×10^{-5} over 2 ms, against 5.4×10^{-5} for the monolithic mesh at the same step. Halving the step reduces both variations by a factor of 3.9 and 4.0 (Table 6, Fig. 3): the variation is the Δt^2 error of central difference, not interface work. The two signs differ because the two discrete systems have different modified energies.

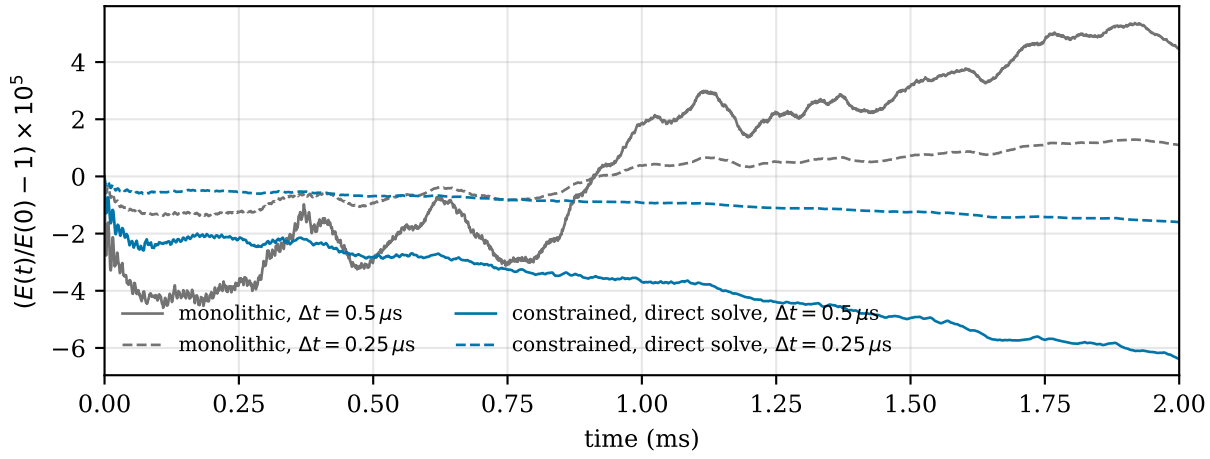


Figure 3: Energy of the neohookean cylinders, −35 m/s, 2 ms: the monolithic mesh and the decomposed mesh with the constrained exchange and the direct interface solve, each at two time steps.

Table 6: Neohookean cylinders, −35 m/s, central difference, 2 ms: $E/E(0)$ at 0.25, 0.5, 1, and 2 ms and the largest $|E/E(0) - 1|$, for the monolithic mesh and the decomposed mesh with the velocity constraint and the direct interface solve (17), at two steps.

system	Δt (s)	0.25 ms	0.5 ms	1 ms	2 ms	max $\lvert\delta E\rvert$
monolithic	5×10^{-7}	0.999963	0.999971	1.000019	1.000045	5.4×10^{-5}
monolithic	2.5×10^{-7}	0.999987	0.999991	1.000004	1.000011	1.4×10^{-5}
decomposed, direct	5×10^{-7}	0.999978	0.999972	0.999963	0.999936	6.4×10^{-5}
decomposed, direct	2.5×10^{-7}	0.999995	0.999993	0.999991	0.999984	1.6×10^{-5}

5.7 Inexact subdomain solves

Every Schwarz iteration solves each subdomain to the Newton tolerance of its solver, although only the converged Schwarz iterate is used. Inexact Schwarz solves the subdomains loosely and relies on the Schwarz iteration, stopped at a tight ϵ, for the accuracy. Its effect was measured on the beam of Section 7 with a neohookean material of the same moduli, so that Newton takes more than one iteration per solve: two trapezoidal subdomains, the constrained exchange with Anderson acceleration, $\epsilon = 10^{-12}$, 10^4 stops of 1 μs on the mesh of level 1 (180 elements) and 2×10^4 stops of 0.5 μs on the mesh of level 2 (1440 elements). The reference run has the relative Newton tolerance 10^{-12}; the error of a run is $\|u - u_{\text{ref}}\|_2 / \|u_{\text{ref}}\|_2$ over all nodes at 10 ms. Table 7 gives the Newton iterations of each run relative to the reference, the Schwarz iterations per stop, and the error, for constant Newton tolerances, for one Newton iteration per solve carried forward

unconverged, and for a forcing term that sets the Newton tolerance of Schwarz iteration k to $\min(\max(\eta\, r_{k-1}, 10^{-12}), 10^{-2})$, with r_{k-1} the relative Schwarz residual of the previous iteration of the stop and 10^{-2} at the first.

Table 7: Neohookean beam, two trapezoidal subdomains, constrained exchange, $\epsilon = 10^{-12}$, 10 ms: Newton iterations relative to the reference (Newton tolerance 10^{-12}), Schwarz iterations per stop, and the relative displacement error against the reference at 10 ms, on the meshes of levels 1 and 2.

Newton setting	level 1, 10^4 stops			level 2, 2×10^4 stops		
	Newton	Schwarz	error	Newton	Schwarz	error
tolerance 10^{-10}	0.930	7.97	1.7×10^{-10}			
tolerance 10^{-6}	0.667	7.97	6.0×10^{-9}	0.670	7.93	3.1×10^{-9}
tolerance 10^{-4}	0.663	7.97	1.7×10^{-5}	0.640	7.93	4.8×10^{-5}
one Newton iteration	0.334	7.97	5.3×10^{-3}	0.335	7.93	3.9×10^{-3}
forcing term, $\eta = 0.01$	0.735	8.52	1.4×10^{-12}	0.717	8.27	1.8×10^{-12}
forcing term, $\eta = 0.1$	0.841	10.18	3.4×10^{-12}	0.826	9.98	2.5×10^{-12}

A constant Newton tolerance does not change the Schwarz iteration count, and its error equals the Newton error, not the Schwarz stopping error. The reason is the structure of the iteration: every Schwarz iteration restarts the subdomain solves from the state at the start of the stop, not from the previous Schwarz iterate, so with a fixed Newton setting each Schwarz iteration applies the same map and the iteration converges to the fixed point of that map, which carries the Newton error. The Schwarz criterion measures the change between iterates and does not see it: with one Newton iteration per solve the iteration meets $\epsilon = 10^{-12}$ in 8 iterations per stop and the result is wrong by 0.5%. The forcing term removes the error at 0.72 to 0.74 of the Newton iterations of the reference. The larger factor, $\eta = 0.1$, raises the Schwarz count by 26 to 28% and loses most of the saving. A tight final solve, which accepts Schwarz convergence only at an iteration whose Newton tolerance was 10^{-12}, was triggered at 50 of the 10^4 stops with $\eta = 0.01$ and did not reduce the error. On a machine with no other load the wall time of the level 1 runs relative to the reference (1081 s on 4 threads) is 0.49 for one Newton iteration per solve, 0.74 for the tolerance 10^{-4}, and 0.81 for $\eta = 0.01$, within 2.4% of the ratios of residual and tangent assemblies. The energy of the neohookean beam under the trapezoidal rule falls by 5.0% over the 10 ms at both levels, identically in the monolithic run; the loss is divided by 3.6 to 3.9 at each halving of the step on a fixed mesh, so it is the second-order error of the integrator at the Courant number the levels share, and the energy is not a measure of the coupling for this material.

On a static problem the picture differs. For a neohookean inclusion in a neohookean matrix (`examples/nonoverlap/static-same-step/matrix-inclusion` at 5% strain in 10 load steps, the classical exchange with Aitken secant relaxation, $\epsilon = 10^{-12}$) Newton tolerances from 10^{-6} to 10^{-10} save 15% of the Newton iterations at an error of 3×10^{-13}, whereas 10^{-4} and looser raise the Schwarz count from 36 to 84 or more per load step and reach the cap of 128, and the forcing term with $\eta = 0.1$ saves 24% at an error of 6×10^{-13}. The classical form of inexact Schwarz, in which each subdomain solve starts from its previous Schwarz iterate so that one Newton iteration per Schwarz iteration converges to the solution of the coupled problem, has not been implemented; it requires the predictor state of the integrator to be carried across Schwarz iterations.

5.8 The classical exchange

The classical Dirichlet–Neumann exchange departs from the constrained problem in the five respects listed in the abstract: it transmits the static reaction $-\boldsymbol{f}^{\text{int}}$, builds its two transfers independently,

imposes displacement, velocity, and acceleration together, stops on the displacement update of the whole domain, and computes the initial acceleration per subdomain with the interface held fixed. The relative difference between the force it applies on the Neumann side and the transposed d'Alembert reaction of the Dirichlet side, measured in the $\boldsymbol{W}_N^{-1}$ norm at the end of every stop, has the median 0.83 (maximum 4.2) with explicit subdomains and 1.47 (maximum 4.6) with implicit ones; under the constrained exchange it is below 4×10^{-16}. On the beam the classical exchange inverts an element within 1 ms with implicit subdomains on every mesh ratio and with 4:1 steps, and with explicit subdomains the energy varies by 2.6 to 4.3% (Fig. 1). On the neohookean cylinders it stays within 3% over 2 ms with explicit subdomains.

The overlap decomposition admits no construction of this kind: its two sides exchange data on two different surfaces, so there is no shared interface on which the two works can cancel. Robin and impedance transmission conditions on a shared interface were measured in the previous version of this note; none conserved on nonconforming meshes, because the transmission condition, not only the Schwarz iteration, defines their fixed point. A Robin condition assembled from the same reaction and the same one-sided constraint has the fixed point of the constrained problem and could serve as a solver of it.

6 Limitations and open questions

(i) *Schwarz stopping error.* Every iterated run stops at a tolerance, and the stopping error accumulates: the conserved quantity drifts by about 10^{-10} over 10^4 stops at $\epsilon = 10^{-12}$, and by about 10^{-9} with the fine-step side as the Dirichlet side. How the drift scales with ϵ and with the number of windows, and whether the criterion (14) needs refinement, has not been measured.

(ii) *Warm start of the subdomain solves.* Each Schwarz iteration restarts the subdomain solves from the state at the start of the stop (Section 5.7), which bounds the saving of inexact subdomain solves at a fixed accuracy to the fraction of Newton iterations beyond the second. Starting each solve from the previous Schwarz iterate would let one Newton iteration per Schwarz iteration converge to the coupled solution; its effect on the Schwarz count and on the pseudo-energy identity has not been measured.

(iii) *Mixed pair with an explicit coarse-step side.* The coarse-step side must be the Dirichlet side and an explicit Dirichlet side has a gain down to -13; Anderson acceleration converges at equal steps in this configuration but has not been measured with subcycling.

(iv) *Cause of the energy swing with the fine-step side as the Dirichlet side.* The swing coincides with an impulse residual of the order of the impulse; that the residual causes it has not been tested separately.

(v) *Nonlinear materials.* Conservation of E to roundoff needs an energy-momentum integrator and a midpoint velocity constraint.

(vi) *Higher flux order and window solves.* Flux order $r \geq 2$ in the sense of Connors et al. and a direct interface solve over a subcycled window are not needed for conservation; whether they reduce the cost has not been measured.

(vii) *Implicit subdomains on the cylinders* and more than two subdomains have not been run.

7 Implementation in Norma

The exchange is a `Schwarz DN nonoverlap` condition with `constrained: true` on both sides of a pair and `constraint: velocity` (or `displacement` for trapezoidal pairs); `default BC type` assigns the Dirichlet and Neumann roles; `interface solve: direct` selects the direct solve for two explicit sides at equal steps. The top-level input sets the relative tolerance ϵ, `relaxation: anderson` with `anderson depth` and `relaxation parameter` (the mixing parameter β, default 0.5 under Anderson acceleration and 1 otherwise), or `relaxation: aitken recursive`; `constraint absolute tolerance` sets an absolute floor in the units of the constrained quantity (default 0), and `stalled interface jump action` selects what a stalled stop does. The block `inexact Newton` (`forcing factor`, default 0.01; `minimum tolerance`, 10^{-12}; `maximum tolerance`, 10^{-2}; `tight final solve`, false) sets the Newton tolerance of the implicit subdomains from the previous Schwarz residual as in Section 5.7. `blended energy output: true` writes, per stop, the energy E (`total_energy`), the pseudo-energy $\tilde{E}$ (`pseudo_energy_total` and per subdomain; NaN for nonlinear materials), the interface displacement and velocity jumps, and, for subcycled pairs, the impulse residual (13). The keys are documented in `docs/src/reference/multidomain.md`. The implementation is in `src/schwarz.jl`, `src/simulation.jl`, and `src/overlap_energy.jl`; tests 135 (pseudo-energy) and 136 (constrained exchange, including the direct solve, subcycling, Aitken and Anderson acceleration, and the equivalence of Anderson acceleration with GMRES on a probed interface map), and 137 (windowed stops). Examples: `examples/nonoverlap/dynamic-same-step/cantilever-dn` (the beam, from which test 136 builds its constrained pairs), `examples/nonoverlap/dynamic-same-step/cantilever-dn-nonconforming` (the beam at mesh ratio 1:0.5), and `examples/jmp/concentric-cylinders-linear`. The parameter study `studies/dynamic-stability` measures the effect of each factor (exchange, mesh ratio, tolerance, time steps, solver, refinement) on the beam.

Data provenance

The equal-step, subcycled, gain, and solver measurements were made on sirius between September 30 and October 1, 2026, at the commits that introduced each feature (`9b09793c` through `570557c5`); the figures and the neohookean time-step measurements are reruns of October 2, 2026 at commit `08b0d866`. The inexact subdomain solve measurements (Section 5.7) were made on sirius (level 1, the inclusion) and rigel (level 2) on October 11 and 12, 2026, at commit `9428ecbd` with the `inexact Newton` control of commit `106800d9`; their data are in `~/Projects/Norma-Schwarz-Coupling/inexact-2026-10`.

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